

Last Time:

□ trajectory forecasting

lecture 11

4R1, FALL '26

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This Time:

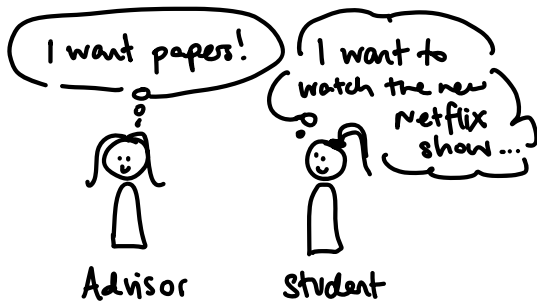
□ game theory

Resources:

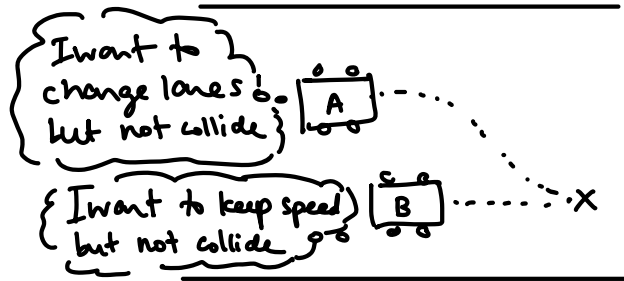
- Tamer Başar and Geert Jan Olsder. "Dynamic noncooperative game theory." SIAM, 1998
- Rufus Isaacs. "Differential Games I: Introduction." RAND Corporation, 1954
- David Fridovich-Keil. "Smooth Game Theory". 2024 .

Why game theory?

A mathematical language for describing "coupled" decision-making problems. ex.



⇒ preferences conflict!
so each actor will choose
strategy which in some
way accounts for the other



⇒ not full of conflict! e.x. both
cars want to cooperate on
the collision part...

Time is also key in games. For example, in chess, each player knows they will get to move in the future.

Dynamic games of this type are particularly interesting in AI / Robotics / HRI.

Terminology

"multi-agent RL"

↗ "game theory"

- agents (or players) - who is playing the game.
- policies (or strategies) - how the players choose their decisions

One may categorize games across several key axes:

- finite / infinite - a game is "finite" if players have only finitely many actions to pick from (e.g. $A = \{\leftarrow, \rightarrow, \uparrow, \downarrow\}$). "Infinite" if the actions available form a continuum (e.g. $A \subseteq \mathbb{R}^n$)

- static / dynamic / differential -

↳ "static" if played @ a single instant in time
(ex. one round of rock-paper-scissors)

↳ "dynamic" if play continues over a period of time
(ex. chess)

↳ "differential" if it is played in continuous time
(e.g. you are optimizing player A and B's trajectories
which are solutions to dynamics ODE
 $\dot{x}_A = f(x_A, u_A)$ and $\dot{x}_B = f(x_B, u_B)$)

- zero / general sum -

↳ "zero sum" are games in which players' objectives add to zero (i.e. your win is my loss). These model perfectly adversarial problems

↳ "general sum" are games with arbitrary player objectives
(e.g. autonomous driving)

- unconstrained / constrained - some games have additional constraints on players' actions or game state.

- pure / mixed strategies -

↳ "pure" strategies are deterministic. Fundamental game theory has shown that not all games have equilibria when players are restricted to pure strategies

↳ "mixed" strategies are stochastic

→ e.g. in rock-paper-scissors, mixed strategies are optimal!

The most relevant type of game we will see in this class is:

{ "infinite, dynamic, general-sum, constrained" games
with "pure" strategies

How do we formulate games?

Let's start simple with static, finite, pure strategy games.

Famous ex: Prisoners' Dilemma!

2 prisoners suspected of crime. Prisoners are guilty, but police need confession b/c they don't have enough evidence. Police tell each prisoner they have 2 options: {confess (C), or quiet (Q)}

Since each player has these options, there are 4 outcomes:

CC: Both confess $\rightarrow$ both given 2 yr. sentence

QQ: Both quiet $\rightarrow$ both given 1 yr. sentence

CQ/QC: The one that confesses gets 0 jail time but the other gets 3 yr. sentence.

Prisoner 1's sentence:

P1 \ P2	C	Q
C	2	0
Q	3	1

convert into
table of
outcomes
for both

Prisoner 2's sentence:

P2 \ P1	C	Q
C	2	3
Q	0	1

$$M_1 := \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix}$$

$$M_2 := \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$$

Imagine P1 decides to confess (C) and P2 decides to stay quiet (Q). We can encode these decisions—or "actions"—as vectors:

$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

x_i is P_i 'th player

action to confess (since its 0 it means you don't)
action to quiet (since its 1 it means you do)

we can write the outcome of the game from each players' perspective as:

$$J_i(x_1, x_2) = x_1^T M_i x_2$$

Annotations:
 - J_i : POV of player $i \in \{1, 2\}$
 - x_1, x_2 : plug in each agents' decision
 - M_i : evaluate "cost" with player i's cost matrix

Each player is choosing $x_1, x_2 \in \underbrace{\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}}_{\text{Action space } A}$. Can we compute best strategy?

Well, if P1 does NOT know what P2 will do, how can it obtain minimal cost? Well, it can play a security strategy which solves:

$$\arg \min_{x_1} \left(\max_{x_2} x_1^T M_{\textcircled{1}} x_2 \right)$$

Annotation: $M_{\textcircled{1}}$ is from Player 1's POV

Ⓚ What order of play is encoded above? i.e. what information does each player know when choosing a strategy?

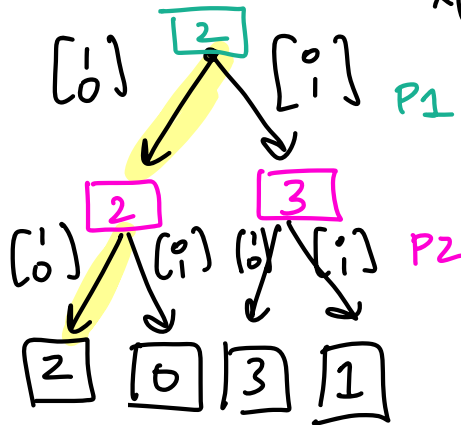
ⓐ Read from left to right. Player 1 first selects a strategy x_1 . Then, Player 2 chooses a response x_2 which maximizes $x_1^T M_1 x_2$ given knowledge of x_1 !

⊗ called security strategy b/c it minimizes P1's cost even when P2 is playing adversarially

ex. (continued)

$$\operatorname{argmin}_{x_1 \in A} \left(\max_{x_2 \in A} x_1^T M_1 x_2 \right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$\begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix}$



which corresponds to worst-case $x_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Q Is it better to play first or second in a zero-sum game?

A Better to play second + exploit knowledge of first player's decision, than 1st and blindly commit to strategy.

Okay, so how do we actually solve games?

Well here, it's important to understand that the solution

— also called an equilibrium — can broadly take

two forms for general-sum games:

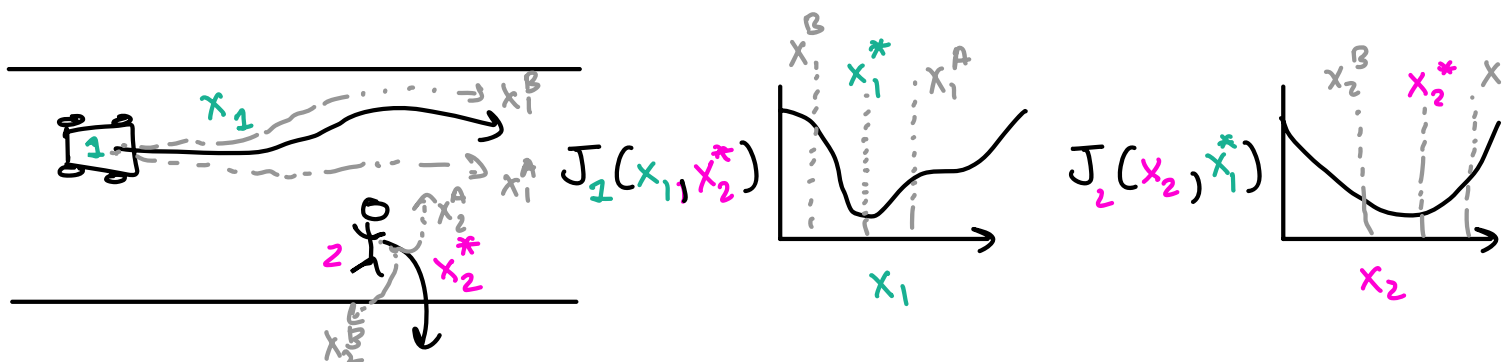
→ after Nobel Laureate John Nash

- Nash Equilibrium (NE) of a 2 player game is a pair of strategies (x_1^*, x_2^*) s.t.

$$J_i(x_i^*, x_{-i}^*) \leq J_i(x_i, x_{-i}^*) \quad \forall x_i \in X_i$$

for player i , the cost of x_i^* (when holding other player constant) is lower or equal

... to any alternative $x_i \in X_i$.



ex (continued, Prisoners' dilemma)

$$M_1 = \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix} \quad M_2 = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$$

The pure strategies $x_1^* = x_2^* = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ form the unique Nash equilibrium of the game $\Rightarrow$ BOTH prisoners confess!

⊛ NOTE: in this case, it happens that x_1^* and x_2^* are also security strategies!

- Stackelberg Equilibrium (SE) which can be seen as the general-sum extension of security strategies.

⇒ Stackelberg equilibria occur when one player must commit to a strategy that the other views before deciding its strategy. The player who pre-commits is called "leader," other is "follower."

$\Rightarrow$ formally a SE is a point (x_1^*, x_2^*) s.t.

$$(P1, \underline{\text{leader}}) \quad x_1^* = \arg \min_{x_1} J_1(x_1, x_2^*)$$

(Pz, follower)

s.t. $x_2^* = \underset{x_2}{\operatorname{argmin}} J_2(x_1, x_2)$

$\underbrace{\hspace{10em}}$

$:= x_2^*(x_1)$

! NOTE: Nash $\neq$ Stackelberg Equilibrium, in general!

How can we algorithmically solve games?

ALGO 0: ITERATED BEST RESPONSE (IBR)

→ this will help us find Nash Equilibria.

Input: initial strategies (x_1, x_2) or $\{x_i\}_{i=1}^N$
start w/ initial "guess"

while not converged do

 for $i = 1, 2 \dots N$

$x_i \leftarrow x_i^*(x_{-i}) := \underset{x_i}{\operatorname{argmin}} J_i(x_i, x_{-i})$ // P_i best response
player i all other players

return converged (x_1^*, x_2^*) ↑ use equilibrium conditions as update rule!

- ⊕ easy to implement w/ standard optimiz. tools
- ⊕ if it converges, it finds NE

- ⊖ no convergence guarantees (cycling!!)
- ⊖ slow convergence $\Rightarrow$ many expensive optimization steps in the inner for loop!

Can we do better?

↳ outside the scope of this class, but there are modern solvers (e.g. PATH*) which can find solutions well + fast!

ex. in Peters et. al. "Contingency Games", RAL 2024
for 5-player, 25 t-step? 3,208 decision vars

$\Rightarrow$ get solution in 35 ms

↳ key idea: search for trajectory that satisfies coupled KKT cond.

OTHER ALGOS:

- Monte Carlo Tree Search
- iterative Linear Quadratic Games (Fridovich-Keil, 2019)

Why is game theory helpful in HRL?

1) joint prediction + planning!

→ recall how before we would call traj. forecasting model to give us $P(X_{H+}^{0:T} | X_H^0)$ and then robot would use predictions to plan $\Rightarrow$ but this doesn't account for influence of R 's actions on H !

2) some settings, games can well-describe interaction phenomena (e.g. economics).